

## PROJECT BRIEF — UNSUPERVISED LEARNING

# Build an AI-Powered Data App

Individual or group project · Practical component

|                    |                               |
|--------------------|-------------------------------|
| WEIGHT             | <b>20% of practical grade</b> |
| TEAM SIZE          | <b>1 – 3 students</b>         |
| PRESENTATION       | <b>15 minutes + Q&amp;A</b>   |
| PRESENTATION DATES | <b>21, 22, 28 and 29 May</b>  |

## What this project is about

You will pick a real-world dataset on a topic you care about, apply an unsupervised learning algorithm to it (clustering or dimensionality reduction), and build a working app that lets someone explore your results. You will present the app live in class.

This is not an exercise in writing code from scratch. The goal is to learn how to build AI-assisted prototypes quickly, use the tools available to you, and communicate what you found in your data, which is something you will need to do a lot in real life.

## Grade impact

**20%**

of practical grade, if you complete and present the project

**80%**

maximum practical grade, if you choose not to participate

Presentation is mandatory to receive credit. There is no alternative submission, the live presentation is the deliverable. If you do not present on your assigned date, you forfeit the 20%.

## What you must deliver

|    |                                           |                                                                                                                                                                                                                             |
|----|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>A dataset</b>                          | Real data on any topic you choose. No synthetic or AI-generated data. It must have enough variables to run clustering or dimensionality reduction on it - please ask the teacher during the class if your dataset is valid. |
| 02 | <b>An unsupervised learning algorithm</b> | Clustering (e.g. K-Means, DBSCAN, hierarchical) or dimensionality reduction (e.g. PCA, t-SNE, UMAP). Apply at least two and visualize them.                                                                                 |
| 03 | <b>A working app</b>                      | Built with React, HTML, or Streamlit. Must display your data and results in a way a non-expert can follow. Does not need to be deployed, running locally on your laptop during the presentation is sufficient.              |

|    |                                      |                                                                                                                                                                                |
|----|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04 | <b>A 15-minute live presentation</b> | Present your app to the class. Explain your dataset, the algorithm you ran, what the results mean, and what you found interesting. Teachers and classmates will ask questions. |
|----|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Rules

- Groups are 1, 2, or 3 students. No exceptions above 3.
- Each group registers their topic with the instructor before starting.
- You are expected and encouraged to use AI coding tools (Cursor, Copilot, ChatGPT, Claude, etc.) to write the code. You are not required to understand every line. You are required to understand what your algorithm does and what your results mean.
- The data must be real. Web-scraped, downloaded from a public repository, or collected by you, all valid. Synthetic data generated by an AI is not valid.
- The app must run live during your presentation.
- Q&A is part of the presentation. Teachers and classmates may ask any question about your data, your method, or your results. You must be prepared to answer, even if you did not write the code yourself.
- Your code must be in a GitHub repository and shared with the instructor ([ivopbernardo](#)) before your presentation date. A private repository is fine, just make sure access is granted in advance.

## Presentation dates

Groups will be assigned to a specific date. Assignments will be communicated in advance. You cannot swap dates without prior approval from the instructor.

|        |                            |
|--------|----------------------------|
| 21 May | Presentations — groups TBD |
| 22 May | Presentations — groups TBD |
| 28 May | Presentations — groups TBD |
| 29 May | Presentations — groups TBD |

## What will be assessed

### App & technical work

- Does the app run and demonstrate the results?
- Is an appropriate algorithm applied to real data?
- Are the results visualised in a way that makes sense?

### Presentation & understanding

- Can you explain what unsupervised learning is doing?
- Can you interpret your clusters or components?
- Can you answer questions from teachers and peers?

## Suggested tools

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### AI coding assistants

Cursor, GitHub Copilot, Claude, ChatGPT, use them freely for code generation.

### Frontend

Streamlit (simplest to start), React, plain HTML/JS.

### ML libraries

scikit-learn (Python) or any equivalent library in your language of choice.

### Data sources

Kaggle, UCI ML Repository, data.gov, any open API, or data you collected yourself.

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Questions about the project must be raised in class or via the course moodle. Topic registration closes two weeks before the first presentation date. Late registration will not be accepted.